

# MindCoach AI Runtime Validator Spec v1.0

**Purpose:** Prevent AI drift. Guarantee MindCoach outputs remain within FoCoCo's locked framework.

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## 0) Inputs and Dependencies

### Authoritative dataset

- MindCoach\_Templates\_v1
- Source of truth must be **Firestore** (mindcoach\_templates) or a versioned JSON file in repo.
- Validator must always validate against the **latest approved schema\_version**.

### AI response (required structured output)

The AI must return **JSON only** with the following keys:

```
{
  "template_id": "MC_T02_PRE_SHOT_FOCUS",
  "routine_type": "📐 Pre-Shot",
  "recommended_cue": "🔊 Trigger Word",
  "delivery_length": "micro_10s",
  "coaching_text": "string",
  "follow_up_question": "string|null"
}
```

If the AI returns anything else (extra keys are okay, missing required keys is not), treat as invalid.

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## 1) Validation Rules (Hard Fail)

### Rule 1: template\_id must exist

- template\_id must match exactly one template in MindCoach\_Templates\_v1.templates[].id
- If not found → **FAIL**

### Fail handling:

Fallback to MC\_T02\_PRE\_SHOT\_FOCUS (safe default) OR “closest match” if you implement mapping. Log the failure.

## **Rule 2: routine\_type must be allowed for that template**

- Look up the template by template\_id
- routine\_type must be in template.allowed\_routine\_types[]

If not → **FAIL**

### **Fail handling:**

Replace routine\_type with the first allowed routine type for that template. Log correction.

## **Rule 3: recommended\_cue must be allowed for that template**

- recommended\_cue must be in template.allowed\_cues[]

If not → **FAIL**

### **Fail handling:**

Replace with safe default cue for that template (recommended order):

1. If allowed cues contains 🧘 Deep Breath → choose it
2. Else choose first element of allowed\_cues  
Log correction.

## **Rule 4: delivery\_length must be allowed for that template**

- delivery\_length must be in template.delivery\_lengths[]

If not → **FAIL**

### **Fail handling:**

Replace with standard\_\* closest available (contains “standard”).

If no standard present, choose first. Log correction.

## **Rule 5: coaching\_text must be present and within safety/tone constraints**

- Must be non-empty string
- Must not exceed max length (recommend):
  - micro: 400 chars
  - standard: 900 chars
  - deep: 1600 chars

If empty / too long → **FAIL**

**Fail handling:**

If too long, trim politely (or re-call model with “shorten” instruction). Prefer trimming for speed.

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## **2) Content Guardrails (Forbidden Language / Claims)**

These are **hard fails** if detected in coaching\_text or follow\_up\_question.

### **A) Medical / therapeutic / diagnostic framing**

Reject if it includes language suggesting diagnosis, treatment, therapy, or mental health care.

**Examples to block (case-insensitive):**

- “diagnose”, “diagnosis”, “treat”, “treatment”
- “therapy”, “therapist”, “counseling”, “counsellor”
- “clinical”, “disorder”, “condition”
- “depression”, “anxiety disorder”, “ADHD”, “OCD” (in MindCoach context)
- “medical”, “psychiatric”, “psychological” (also blocked per your brand rules)
- “I can’t replace your doctor/therapist” (even this is therapy-adjacent; keep it out of MindCoach)

**Fail handling:**

Replace coaching\_text with a safe generic version:

- Performance-only, golf-only language
- Re-run generation with strict instruction: “golf performance only, no health language”

### **B) Overpromising outcomes**

Reject if it guarantees results.

Block phrases like:

- “This will fix...”
- “You will definitely...”
- “Guaranteed...”
- “Cure...”

**Fail handling:**

Replace with probabilistic, practical language:

- “This helps you...”
- “Try this to...”
- “The goal is...”

**C) Template invention / framework drift**

Reject if it introduces:

- new “modes”, “systems”, “pillars”, “cues”, “steps” framed as new named methods outside dataset

Signals:

- “New cue: ...”
- “Add this routine type: ...”
- “Fourth pillar...”

**Fail handling:**

Strip invented names, map guidance back into allowed cues and template structure, or re-run.

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**3) Optional Validation (Recommended but Not Hard Fail)****A) follow\_up\_question rules**

- Allowed to be null/empty
- If present:
  - Must be a single question
  - Must be short (max 140 chars)
  - Must not ask sensitive health questions
  - Must stay golf-contextual

If violated: set to null and log.

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**4) Logging Requirements (Non-negotiable)**

Every MindCoach response must create a validator log record.

## **Collection: ai\_validator\_logs**

Fields:

- timestamp
- user\_id
- template\_id\_requested (if any)
- template\_id\_returned
- validator\_status = PASS | FAIL\_CORRECTED | FAIL\_FALLBACK
- failed\_rules[] (e.g., ["template\_id\_not\_found", "cue\_not\_allowed"])
- replacements (object showing old → new values)
- model\_version (LLM version)
- prompt\_version (your system prompt version)
- content\_flags (e.g., ["medical\_language\_detected"])

This gives you governance + auditability.

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## **5) Runtime Flow (Implementation Steps)**

### **Step 1: Determine context (pre-AI)**

Inputs might include:

- user location in app (MindCoach, During Round, Post Shot)
- current routine type
- last logged mindset rating
- pressure flag (user-selected or inferred)
- last template used (avoid repeating too often)
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### **Step 2: Choose template\_id (either system chooses or AI suggests)**

**Recommended approach:** system chooses template\_id first.  
Then AI only generates within it.

### Step 3: Generate AI response with strict schema

- Provide:
  - selected template object
  - required output schema
  - constraints list
  -

### Step 4: Validate response

Apply rules 1–5 + guardrails.

### Step 5: If FAIL → correct or fallback

- If minor mismatch (cue/routine/length): correct fields
- If content guardrail violation: regenerate or fallback to safe generic guidance

### Step 6: Save session + validator log

- Save to mindcoach\_sessions
- Save validator log

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### 6) Fallback Safe Responses (Pre-written)

Use these when content fails repeatedly or model output is malformed.

#### Fallback A: Pre-Shot Focus (safe)

- template\_id: MC\_T02\_PRE\_SHOT\_FOCUS
- routine\_type:  Pre-Shot
- recommended\_cue:  Deep Breath (if allowed; else first allowed cue)
- coaching\_text:  
“One breath. Pick one target. Commit to one swing. Then go.”

#### Fallback B: Mistake Recovery (safe)

- template\_id: MC\_T05\_MISTAKE\_RECOVERY
- routine\_type: 🏃 Between Shots
- recommended\_cue: 🔄 Reset
- coaching\_text:  
“Reset. The last shot is done. Your job now is one clear decision for the next one.”

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## 7) Versioning Rules

- Store:
  - schema\_version and dataset name in every template doc
- AI prompts must reference:
  - MindCoach\_Templates\_v1
- Any edits create MindCoach\_Templates\_v1.1 (or v2), never overwrite silently.